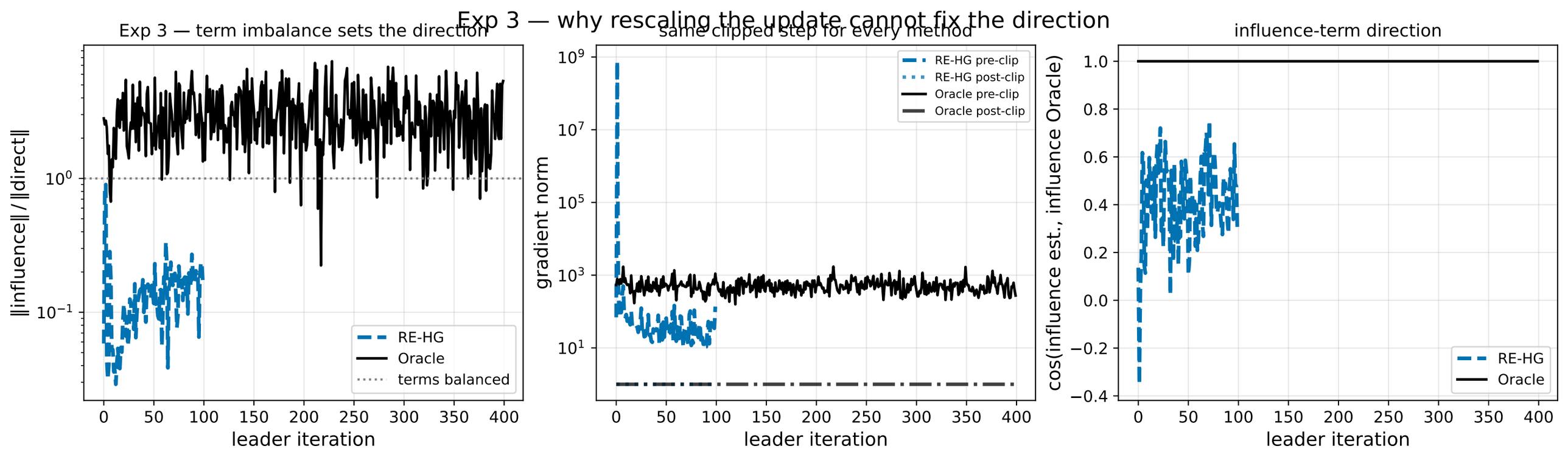
Exp 3 — why rescaling the update cannot fix the direction
Exp 3 — term imbalance sets the direction
‖influence‖ / ‖direct‖
10^0
10^{-1}
RE-HG
Oracle
terms balanced
0
50
100
150
200
250
300
350
400
leader iteration
same clipped step for every method
gradient norm
10^9
10^7
10^5
10^3
10^1
RE-HG pre-clip
RE-HG post-clip
Oracle pre-clip
Oracle post-clip
leader iteration
influence-term direction
cos(influence est., influence Oracle)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
RE-HG
Oracle
leader iteration